Quantitative Macro Theory **HW #1**
Spring 2018 **Due at 3pm on Wed 1/24/2018**

Write your answer on latex and prepare a pdf file. The file should be clearly written and organized. You should think of that file as a short version of a paper. The appendix of that file should include your code. Email me that pdf file before the deadline.

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1. (50 points) Consider a deterministic version of the neoclassical growth model:

$$\max_{c_t, k_{t+1}} \sum_{t=0}^{\infty} \beta^t \frac{c_t^{1-\sigma}}{(1-\sigma)},$$

subject to,

$$c_t + k_{t+1} = A k_t^{\theta} + k_t(1-\delta).$$

   (a) (15 points) Write down the Euler equation for this model. Set $\sigma = 2$, $\beta = 0.9$, $\theta = 1/3$, $\delta = 0.07$, and $A = (\beta^{-1} + \delta - 1)/(\theta)$. Set $k_0 = 0.85$ and imagine you know that $k_2 = 0.95$. Find the value of k_1 that solves the Euler equation.

   (b) (25 points) Starting at $k_0 = 0.75$ find the path of k_t from $t = 0$ until $t = 200$ using the extended path method I described in class. Starting from your guess $\{k_t^0\}$, you will have a revised path for k_t in each iteration j, i.e., $\{k_t^j\}$. Plot the paths for $j = \{0, 1, 2, 5\}$ together with the optimal path.

   (c) (10 points) Plot the paths for investment, consumption, and the real interest rate from $t = 0$ until $t = 200$. Interpret the results.

2. (50 points) Let's use the same model that in the previous question. But instead of constant productivity, consider

$$A_t = A \times \left(1 + 0.05 \times \left(1 - \frac{t^{1/2}}{7}\right)\right) \text{ for } t = 1, ..., 49.$$

and $A_t = A$ for $t = 50, ..., 250$.

   (a) (40 points) Starting at $k_0 = k^*$, find the optimal path of k_t from $t = 0$ until $t = 250$ using the extended path method I described in class. Plot the paths for k_t that you obtained in iteration $j = \{0, 1, 2, 5\}$ together with the optimal path.

   (b) (10 points) Plot the paths for investment, consumption, and the real interest rate from $t = 1$ to $t = 50$. Interpret the results.